Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)	KE + gravitational PE = 0 & **context explained briefly** (also accept KE = gravitational PE and context) (1) Minimum context = conservation of energy **or** mention of galaxy **or** spherical universe **or** uniform universe **or** cosmological principle **or** explaining why $r = D$ or explaining what the mass of the sphere is (accept whole universe or part and mass inside) Reference to universe nor expanding now loses 1st mark $v = H_0D$ substituted for v (1) mass = density × volume of sphere (1) Algebra set out clearly (1)	4			4	3	
		(ii)	Correct substitution into formula (gives 8.66×10^{-27}) (1) Dividing by 1.66×10^{-27} (or 5.2 or 5.22 seen) OR 5 u = 8.3×10^{-27} (1)	1	1		2	1	
	(b)	(i)	Halfway between or correct substitutions into C of M equation leading to 1×10^{11} or simply 1×10^{11} [m]		1		1		
		(ii)	Substitution into period equation (1) 9.73×10^6 s (allow 1 mark for 13.8×10^6 OR 4.87×10^6 OR 3.44×10^6) (1)	1	1		2	2	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(iii)	Use of $v = \frac{2\pi r}{T}$ ecf **or** equivalent with $v = \omega r$ and $\omega = \frac{2\pi}{T}$ (1) (expect 65 000 m s^{-1}) Use of Doppler equation (1) Correct answer = 0.0934 nm (accept 1sf) (1) Note: ecf on previous values gives 0.187 nm, 0.264 nm and 0.066 nm (for full marks unless another clear mistake seen) Allow 2 marks max if d used instead of r (gives 0.187 nm) Allow 1 mark max if $\frac{1}{2}mv^2 = \frac{GMm}{r}$ (also gives 0.187 nm)	1 1	1		3	3	
	(c)		Centre of mass is nearer more massive star (1) Which will give a larger field OR but must be nearer smaller star for fields to match (1) Accept equivalent reverse argument		2		2	2	
	(d)	(i)	Force [due to both stars] are always equal [and opposite] Accept this is the point where the field is zero Accept masses of equal **mass and distance** either side		1		1		
		(ii)	Force due to black hole is 4×greater or 4.17×10^{30} [N] (1) Also force from other star (gives factor 5) or 1.04×10^{30} [N] (1)		2		2	1	

Question			Marking details	AO1	AO2	AO3	Total	Maths	Prac
				Marks available					
		(iii)	Increasing v increases redshift **OR** linking orbital speed to redshift (can be implied by good answers) (1) Correct realisation that factor is $\sqrt{5}$ or that v^2 proportional to force (this implies the 1st mark) (1) Joseff is wrong (based on some valid physics) (1) e.g. red shift increases with orbital speed but they aren't proportional (force and speed) so Jo is wrong (this gets 2 marks – 1st and 3rd) **Also accept for 2 marks:** Red shift increases with orbital speed and orbital speed is proportional to force, so Jo is right **Also accept calculations:** Expected v = 144 000 m s^{-1} (1) Red shift = 0.21 n[m] (1) Joseff is wrong (1)			3	3	2	
	(e)		[More] experiments involving Higgs [decay] (1) Results [need to be] in agreement with theory (1)			2	2		
			Question 5 total	**8**	**9**	**5**	**22**	**12**	**0**